SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Name :- R. Jaideep
Reg No :- 192505441

Programme: **B.E / B.Tech**
Course Code: **MLA02**
Course Name: **FUNDAMENTALS OF MACHINE LEARNING**
Duration: **1:30 Hr**
Date : 31/10/2026 .

Max Marks: **50 Marks**

ASSIGNMENT – I

UNIT I – INTRODUCTION AND MATHEMATICAL FOUNDATIONS

Topics: *Design of a learning system, types of machine learning, probability theory and distributions, Bayes decision theory, information theory*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO1	Analyse machine learning fundamentals and mathematical foundations, including probability theory, Bayes decision theory, and information theory.

Answer all the Tasks

Task 1. A retail bank plans five machine-learning systems. For each system, analyse the problem and state (i) the learning paradigm (supervised / unsupervised / semi-supervised / reinforcement), (ii) the learning task (classification, regression, clustering, control, etc.) and (iii) the experience E and performance measure P in Mitchell's (T, P, E) formulation:

(a) Predict whether a loan applicant will default, using ten years of labelled repayment records.

(b) Discover behavioural customer segments among 2 million customers for which no labels exist.

(c) Estimate the next-month credit-card spend (in ₹) of each customer.

(d) Learn an ATM cash-refill policy that maximises cash availability minus transport cost through repeated trial and feedback.

(e) Only 2% of 500,000 transaction records are labelled fraud/genuine; the rest are unlabelled. Build a fraud detector that exploits both kinds of data.

Finally, justify which of the five systems is most harmed by label noise.

(5) (CO1) (BL4)

Task 2. A screening test for a disease with prevalence 2% has sensitivity 95% and specificity 90%. Calculate the probability that a person who tests positive actually has the disease, and analyse why the answer is much lower than the sensitivity.

(2) (CO1) (BL4)

Task 3. The joint probability mass function of a discrete random vector (X, Y) is given below.

X \ Y	Y = 0	Y = 1
X = 0	0.10	0.20
X = 1	0.15	0.25

X \ Y	Y = 0	Y = 1
X = 2	0.20	0.10

Analyse the distribution by (a) finding the marginals of X and Y, (b) computing E[X], Var(X) and E[Y], (c) computing $P(X = 2 \mid Y = 1)$, (d) computing Cov(X, Y) and the correlation coefficient, and (e) deciding whether X and Y are independent, with a justification.

(6) (CO1) (BL4)

Task 4. A temperature sensor output is modelled as $X \sim N(\mu = 50, \sigma^2 = 16)$.

(a) Calculate $P(X > 58)$ and $P(46 < X < 54)$.

(b) A second sensor gives the readings 12, 15, 11, 14, 13. Derive the maximum-likelihood estimates of its mean and variance and compare the ML variance with the unbiased estimate, stating which is biased and why. (Use $\Phi(1) = 0.8413$, $\Phi(2) = 0.9772$.)

(4) (CO1) (BL4)

Task 5. A source emits four symbols with probabilities (1/2, 1/4, 1/8, 1/8). Calculate its entropy, compare it with the entropy of a uniform source over four symbols, and analyse the redundancy. Also compute the average length of the code {0, 10, 110, 111} and comment on its optimality.

(3) (CO1) (BL4)

Task 6. A model predicts class probabilities Q = (1/3, 1/3, 1/3) while the true distribution is P = (0.5, 0.25, 0.25). Compute KL(P‖Q) and KL(Q‖P) in bits, and analyse what the asymmetry implies when KL divergence is used as a training loss.

(3) (CO1) (BL4)

Task 7. Two binary variables A and B have the joint distribution $P(0,0) = 0.4$, $P(0,1) = 0.1$, $P(1,0) = 0.1$, $P(1,1) = 0.4$. Calculate H(A), H(B), the joint entropy H(A,B), the conditional entropy H(A|B) and the mutual information I(A;B). Analyse what the value of I(A;B) says about using B as a feature to predict A.

(4) (CO1) (BL4)

Task 8. A one-dimensional feature x follows $p(x|\omega_1) = N(2, 4)$ for class ω_1 and $p(x|\omega_2) = N(6, 4)$ for class ω_2, with priors $P(\omega_1) = 0.6$ and $P(\omega_2) = 0.4$ and zero–one loss. Derive the Bayes decision threshold x^*, classify $x = 4.5$, and calculate the overall probability of error. Analyse how the threshold moves as the prior of ω_1 increases.

(4) (CO1) (BL4)

Task 9. For an observation x, the posteriors are $P(\omega_1|x) = 0.3$ (disease) and $P(\omega_2|x) = 0.7$ (healthy). Action α_1 = 'declare disease', α_2 = 'declare healthy'. The loss matrix is $\lambda(\alpha_1|\omega_1) = 0$, $\lambda(\alpha_1|\omega_2) = 8$ (unnecessary treatment), $\lambda(\alpha_2|\omega_1) = 50$ (missed disease), $\lambda(\alpha_2|\omega_2) = 0$. Evaluate the conditional risks, decide the minimum-risk action, compare it with the MAP decision and determine the posterior $P(\omega_1|x)$ above which α_1 is optimal.

(3) (CO1) (BL5)

Task 10. Polynomial regression models of increasing degree were fitted to the same dataset, giving the mean-squared errors below.

Degree	**1**	**2**	**3**	**5**	**9**
Training MSE	9.0	4.1	2.2	0.9	0.1

Degree	1	2	3	5	9
Validation MSE	9.5	4.6	2.9	3.6	11.8

Calculate the generalisation gap for each degree, diagnose under-fitting/over-fitting using the bias–variance trade-off, select the best model with justification, and recommend two techniques to improve the degree-9 model.

(4) (CO1) (BL5)

Task 11. A coin is tossed 10 times and shows 7 heads. The bias θ = P(head) is given a Beta(2, 2) prior. Calculate the ML estimate, the MAP estimate and the posterior mean of θ. Then evaluate how the three estimates change if the data were 70 heads in 100 tosses, and explain the role of the prior.

(4) (CO1) (BL5)

Task 12. A hospital wants an early-warning system that raises an alert when an ICU patient is likely to develop sepsis within the next 6 hours. Historical electronic health records (vital signs, lab values, outcomes) are available.

(a) Design the learning system: state the task T, performance measure P and experience E, the target function, a suitable input representation, a learning algorithm and the validation strategy. [3]

(b) On a validation cohort of 1,000 patients, 5% develop sepsis. The trained model has sensitivity 0.90 and specificity 0.80. Compute TP, FN, FP and TN. [1]

(c) Compute the positive predictive value (PPV) and negative predictive value (NPV). [1]

(d) A missed sepsis case costs ₹100 (unit cost) and a false alarm costs ₹5. Compare the total cost of (i) using the model, (ii) alerting every patient and (iii) never alerting. [2]

(e) Recommend a deployment policy and justify how the alert threshold should be set using Bayes decision theory. [1]

(8) (CO1) (BL6)

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ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - I

Programme	B.E / B.Tech	**Course Code**	**MLA02**
Course Name	**Fundamentals of Machine learning.**	**Date**	
Duration	1 Hr 30 Min	**Max Marks**	50

Register No.: 192505441

TASK 1	Max Marks	Marks Awarded
Task 1	5	
Task 2	2	
Task 3	6	
Task 4	4	
Task 5	3	
Task 6	3	
Task 7	4	
Task 8	4	
Task 9	3	
Task 10	4	
Task 11	4	
Task 12	8	
TOTAL	**50**	

Total in words:

47/50

Machine learning

MLA0204

ASSIGNMENT-1

Name:- R. Jaideep

RegNO:- 192525441

Depart:- AI & ML

Task 1. Design of learning Systems.

Mitchell's Formulation

(T, P, E)

T: Task performance by the system

P: performance measure used to evaluate the system

E: Experience or training data used.

System	learning paradigm	learning task.
(a) loan default prediction	Supervised	classification
(b) customer Segmentation	unsupervised	clustering
(c) Credit card spending	Supervised	Regression.

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Label noise analysis:

System (a), loan default prediction, is highly affected by label noise because it depends entirely on labelled payment records. Incorrect labels can make the model learn wrong patterns and produce incorrect Predictions.

Task 2: Probability of Disease Using Baye's theorem

Given;

prevalence = 2% = 0.02

Sensitivity = 95% = 0.95

Specificity = 90% = 0.90

False positive rate = 1 - 0.90 = 0.10

Using Baye's theorem:

$$P(D/+) = \frac{P(+/D)\,P(D)}{P(+/D)\,P(D) + P(+/\bar{D})\,P(\bar{D})}$$

$$= \frac{0.95 \times 0.02}{(0.95 \times 0.02) + (0.10 \times 0.98)}$$

$$= \frac{0.019}{0.117}$$

$$P(D/+) = 0.162$$

$$P(D/+) = 16.2\%$$

∴ The probability is much lower than sensitivity because the disease prevelance is only 2%. Many healthy people can all test positive in false positives.

Task 3. Joint probability Distribution

Given, joint probability distribution

X/Y	Y=0	Y=1
X=0	0.10	0.20
X=1	0.15	0.25
X=2	0.20	0.10

(a) Find Marginal probabilities of X & Y.

Marginal probability of X:

$$P(X=0) = 0.10 + 0.20 = 0.30$$

$$P(X=1) = 0.15 + 0.25 = 0.40$$

$$P(X=2) = 0.20 + 0.10 = 0.30$$

$$P(Y=0) = 0.10 + 0.15 + 0.20 = 0.45$$

$$P(Y=1) = 0.20 + 0.25 + 0.10 = 0.55$$

(b) calculate $E[X]$, $var(x)$ and $E[Y]$

Expected value of x:

$$E[x] = \sum x P(x)$$

$$= 0(0.30) + 1(0.40) + 2(0.30)$$

$$= 0 + 0.40 + 0.60$$

$$E[x] = 1$$

Variance of x

$$var(x) = E[x^2] - [E(x)]^2$$

$$E[X^2] = 0^2(0.30) + 1^2(0.40) + 2^2(0.30)$$

$$= 1.60$$

$$Var(X) = 1.60 - (1)^2$$

$$Var(X) = 0.60$$

Expected value of Y:

$$E[Y] = 0(0.45) + 1(0.55)$$

$$E[Y] = 0.55$$

(c) calculate $P(X=2|Y=1)$

using conditional probability

$$P(X=2|Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)}$$

$$= \frac{0.10}{0.55}$$

$$P(X=2|Y=1) = 0.1818.$$

(d) calculate covariance & Correlation coefficient

$$Cov(X,Y) = E(X,Y] - E[X]E[Y]$$

$$E[XY] = (0\times0\times0.10) + (0\times1\times0.20) + (1\times0\times0.15)$$

$$+ (1\times1\times0.25) + (2\times0\times0.20) + (2\times1\times0.10)$$

$$= 0.25 + 0.20$$

$$= 0.45$$

$$\therefore Cov(X,Y) = 0.45 - (1\times0.55)$$

$$Cov(X,Y) = -0.10$$

Correlation Coefficient

$$r = \frac{Cov(X,Y)}{\sqrt{Var(X)\,Var(Y)}}$$

$$Var(Y) = 0.55(1-0.55)$$
$$= 0.2475$$

$$r = \frac{-0.10}{\sqrt{0.60 \times 0.2475}}$$

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$$\boxed{r = -0.2593}$$

Task 4: Normal Distribution & Maximum likelihood Estimation

(a) calculate the probabilities.

Given,

$$X \sim N(50, 16)$$

Mean (μ) = 50

Standard deviation (σ) = $\sqrt{16} = 4$.

Standard normal Formula $z = \frac{X - \mu}{\sigma}$

1. calculate $P(x > 58)$

$$z = \frac{58 - 50}{4} = 2$$

$$P(x > 58) = P(z > 2)$$
$$= 1 - \Phi(2)$$
$$= 1 - 0.9772$$

$$\boxed{P(x > 58) = 0.0228}$$

2. Calculate $P(46 < x < 54)$

for $x = 46$

$$z_1 = \frac{46-50}{4} = -1$$

For $y = 54$;

$$z_2 = \frac{54-50}{4} = 1$$

$$P(46 < y < 54) = \phi(1) - \phi(-1)$$

$$= 0.8413 - 0.1587$$

$$\boxed{P = 0.6826}$$

(b) Maximum Likelihood Estimation!

Given reading

12, 15, 11, 14, 13

Step 1 : calculate mean

$$\mu = \frac{\Sigma x_i}{n}$$

$$= \frac{12+15+11+14+13}{5} = 13$$

Step 2: calculate ML variance

$$\sigma^2 = \frac{E(x_i - \mu)^2}{n}$$

$$= (12-13)^2 + (15-13)^2 + (11-13)^2 + (14-13)^2 + (13-13)^2$$

$$= \frac{1+4+4+1+0}{5}$$

$$= 10/5$$

$$\boxed{\sigma^2 = 2}$$

Step 3: unbaised variance

$$s^2 = \frac{\Sigma (x_i - \bar{x})^2}{n-1}$$

$$= \frac{10}{5-1}$$

$$\boxed{s^2 = 2.5}$$

ML variance = 2

unbaised variance = 2.5

Task 5 : Information Theory of entropy

Given probabilities

$$P = (1/2, 1/4, 1/8, 1/8)$$

Step 1: calculate entropy

entropy: $H(X) = -\Sigma P(x) \log_2 P(x)$

$$= -\left[\frac{1}{2}\log_2 \frac{1}{2} + 1/4 \log_2 1/4 + 1/8 \log_2 1/8 + 1/8 \log_2 1/8\right]$$

$$= 0.5 + 0.4 + 0.375 + 0.375$$

$$\boxed{H(X) = 1.75 \text{ bit}}$$

Step 2: uniform source entropy

$$H_{max} = \log_2 4$$

$$H_{max} = 2 \text{ bits.}$$

Step 3: calculate redundaly.

$$R = H_{max} - H(x)$$

$$= 2 - 1.75$$

$$\boxed{R = 0.25 \text{ bit}}$$

$$= \frac{0.25}{2} \times 100$$

$$= 12.5\%$$

Step 4: Average code length

given code:

111

code lengths = 1, 2, 3, 3.

$$L = \sum P(x_i) l_i$$

$$= 1/2(1) + 1/4(2) + \frac{1}{8}(3) + \frac{1}{8}(3)$$

$$= 0.5 + 0.5 + 0.375 + 0.375$$

$$\boxed{L = 1.75 \text{ bit}}$$

Task 6: KL Divergence

given

$$P = (0.5, 0.25, 0.25)$$

$$Q = (1/3, 1/3, 1/3)$$

$$D_{KL}(P||Q) = \sum P(x) \log_2 \frac{P(x)}{Q(x)}$$

1. calculate $KL(P||Q)$

$$= 0.5 \log_2 \frac{0.5}{1/3} + 0.25 \log_2 \frac{0.25}{1/3} +$$

$$0.25 \log_2 \frac{0.25}{1/3}$$

$$\boxed{D(KL)(P||Q) = 0.0850 \text{ bit}}$$

2. Calculate $KL(Q||P)$

$$D_{KL}(Q||P) = \sum Q(x) \log_2 \frac{Q(x)}{P(x)}$$

$$= 1/3 \log_2 \frac{1/3}{0.5} + 1/3 \log_2 \frac{1/3}{0.25} + 1/3 \log_2 \frac{1/3}{0.25}$$

$$\boxed{D_{KL}(Q||P) = 0.0817 \text{ bits}}$$

KL divergence is asymmetric

$$D_{KL}(P||Q) \neq D_{KL}(Q||P)$$

Task 7: Entropy and Mutual Information

Given,

$P(0,0) = 0.4$ $P(0,1) = 0.1$

$P(1,0) = 0.1$ $P(1,1) = 0.4$

1. Calculate $H(A)$ & $H(B)$

Marginal probabilities

$P(A=0) = 0.5$, $P(A=1) = 0.1$

$P(B=0) = 0.5$, $P(B=1) = 0.5$

Entropy $H(A) = -\sum P(A) \log_2 P(A)$

$= -0.1 \log_2 0.5 - 0.5 \log_2 0.5$

$\boxed{H(A) = 1 \text{ bit}}$ $\boxed{H(B) = 1 \text{ bit}}$

2. 'Joint' entropy $H(A,B)$

$H(A,B) = -\sum P(A,B) \log_2 P(A,B)$

$= -[2(0.4 \log_2 0.4) + 2(0.1 \log_2 0.1)]$

$\boxed{H(A,B) = 1.7219 \text{ bits}}$

3. Conditional entropy $H(A|B) = H(A,B) - H(B)$

$= 1.7219 - 1$

$\boxed{H(A|B) = 0.7219 \text{ bits}}$

Task 8:- Bayes Decision Theory

Given

$P(x/w_1) = N(2,4)$ $\quad P(w_1) = 0.6$

$P(x/w_2) = N(6,4)$ $\quad P(w_2) = 0.4$

1. Derive Bayes decision threshold

$$P(w_1)\,P(x/w_1) = P(w_2)\,P(x/w_2)$$

$$\ln \frac{0.6}{0.4} = \frac{(x-2)^2 - (x-6)^2}{8}$$

$$\ln(1.5) = x - 4$$

$$x^* = 4 + \ln(1.5)$$

$$x^* = 4.4055$$

2. classify $x = 4.5$

$4.5 > 4.4055$ $\boxed{w_2}$

3. Calculate overall probability of error.

$$P(\text{error}) = P(w_1)\,P(x > x^*/w_1) + P(w_2)\,P(x < x^*/w_2)$$

$\sigma = \sqrt{4} = 2$

$$z_1 = \frac{4.4055 - 2}{2} = 1.2027$$

$$z_2 = \frac{4.4055 - 6}{2} = -0.7973$$

$$P(\text{error}) = 0.6(0.1145) + 0.4(0.2126)$$

$$= 0.0687 + 0.0850$$

$$P(\text{error}) \approx 0.1538$$

15.38%

Task 9:- Minimum Risk classification

$P(w_1/x) = 0.3$

$P(w_2/x) = 0.7$

(1) Calculate Conditional risk

$R(\alpha_1|x) = 0(0.3) + 8(0.7)$

$R(\alpha_1|x = 5.6)$

Risk of declaring health

$R(\alpha_2|x) = 50(0.3) + 0(0.7)$

$\boxed{R(\alpha_2|x) = 15}$

(2) Minimum-risk action

$5.6 < 15$

$\boxed{\alpha_1 = \text{Declare disease}}$

(3) compare with MAP decision

MAP selects the class with the highes posterior probability.

$P(w_2|x) = 0.7$

(4) Find posterior threshold

$8(1-P) < 50P$

$8 < 58P$

$P > 8/58$

$\boxed{P > 0.1379}$

Task 10: Generalisation Gap and Bias-variance Trade-off

given training and validation MSE

Degree	Training MSE	Validation MSE
1	9.0	9.5
2	4.1	4.6
3	2.2	2.9
5	0.9	3.6
9	0.1	11.8

(1) calculate generalisation gap

Formula: Gap = validation MSE − Training MSE

Degree	calculation	Gap
1	9.5 − 9.0	0.5
2	4.6 − 4.1	0.5
3	2.9 − 2.2	0.7
5	3.6 − 0.9	2.7
9	11.8 − 0.1	11.7

2. Analyse underfitting & overfitting

Degree 1 : underfitting because training & validation errors are high

Degree 2 : Better performance with lower errors.

Degree 3 : Good balance b/w bias & variance

Degree 5 : overfitting begins because training error is low

Degree 9 : Severe overfitting because to mode memorizes.

3. Select the Best Model :-

Validation MSE = 2.9

2 techniques improve degree-9 model :-

① Regularisation

② cross-validation

explain.

Task II : Maximum Likelihood, MAP & Bayesian Estimation.

given,

Total tosses = 10, Heads = 7, Tails = 3, prior = Beta(2,2)

1. Maximum Likelihood Estimation (MLE) = $\frac{\text{Heads}}{\text{Total}} = 7/10$

MLE = 0.7

2. Maximum A posteriori (MAP) :- $\alpha' = 2+7 = 9$ $\beta' = 2+3 = 5$

$$MAP = \frac{\alpha' - 1}{\alpha' + \beta' - 2} = 8/12$$

MAP = 0.6667

3. Posterior Mean :- $= \frac{\alpha'}{\alpha' + \beta'} = 9/14 = 0.6429.$

4. For 100 tosses (70 heads, 30 tails) MLE = $70/100 = 0.7$

MAP = $71/102 = 0.6961$

Posterior Mean = $\frac{72}{104} = 0.6923.$

Task_2: Hospital Sepsis Early-warning System.

@ Design the learning System.

→ Task (T) : predict whether an icu patient will develop sepsis within 6 hours.

→ performance (P): Accuracy, sensitivity, specificity and recall.

→ Experience (E): Historical patient records containing vital signs and laboratory results.

→ learning algorithm : Random forest (or) logistic Regression.

→ Output : Sepsis (1) (or) No sepsis (0).

(b) Calculate TP, FN, FP and TN

given

Total patients = 1000

Prevalance = 5%

Sensitivity = 90%

Specificity = 80%

Sepsis patient $1000 \times 0.05 = 50$

Actual healthy patients = 950

$TP = 50 \times 0.90 = 45$

$FN = 50 - 45 = 5$

$TN = 950 \times 0.80 = 760.$

$FP = 950 - 760$
$= 190.$

	Sepsis	+Healthy
Predicted positive	45	190
Predicted negative	5	760

(c) calculate PPV and NPV

Positive predictive value.

$$PPV = \frac{TP}{TP+FP} = \frac{45}{235}$$

PPV = 19.15%

$$NPV = \frac{TN}{TN+FN} = \frac{760}{765} = 99.35\%$$

(d) Compare total costs.

Given, Missed sepsis cost = ₹100
False alarm cost = ₹5

cost = (5 x 100) + (190 x 5) = ₹1450.

Alerting every patient
950 x 5 = ₹ 4750

Never alerting: 50 x 100 = ₹ 5000

(e) Recommend deployment policy.

The hospital should use the model as an early warning system with doctors reviewing alerts.

$5(1-p) < 100p$

$5 < 105p$

$p > 0.0476 = 4.76\%$

The model can help identify high risk patient early, reduce missed sepsis cases & support doctors in decision making.